

Instagram Data Analysis

Alfido Tech Data Science Internship – Task 3

1. Project Overview

This project was completed as part of the Alfido Tech Data Science Internship.

The objective of this project is to perform Exploratory Data Analysis (EDA) on Instagram post data and understand engagement patterns based on post type, likes, and comments.

2. Dataset Overview

Dataset: Instagram Posts Dataset

Number of Posts: 40

Number of Columns: 4

Missing Values: None

Target Metrics: Likes and Comments

The dataset contains information about Instagram posts, including post type, likes, and comments.

3. Tools and Technologies

Python

Pandas

Matplotlib

Google Colab

Exploratory Data Analysis (EDA)

4. Analysis Performed

The following analysis was performed:

Loaded and inspected the Instagram dataset.

Checked the dataset shape and column names.

Checked data types and missing values.

Generated descriptive statistics.

Analyzed the distribution of Instagram post types.

Compared average likes across different post types.

Compared average comments across different post types.

Examined the relationship between likes and comments.

Generated four visualizations.

5. Visualizations

6. Distribution of Instagram Post Types

Bar chart showing the number of posts for each post type.

2. Average Likes by Post Type

Bar chart comparing the average likes received by Carousel, Image, and Reel posts.

3. Comments vs Likes

Scatter plot showing the relationship between comments and likes.

4. Average Comments by Post Type

Bar chart comparing average comments across different post types.

6. Key Insights

The dataset contains 40 Instagram posts.

Carousel is the most common post type.

The overall average number of likes is 18,098.72.

The overall average number of comments is 216.38.

The maximum likes recorded are 79,000.

The maximum comments recorded are 1,089.

Carousel posts have the highest average likes at approximately 20,518.80.

Image posts have an average of approximately 18,543.18 likes.

Reel posts have an average of approximately 15,156.57 likes.

Reel posts have the highest average comments at approximately 316.71.

Carousel posts have an average of approximately 180.53 comments.

Image posts have an average of approximately 137.55 comments.

7. Conclusion

The analysis provides an overview of Instagram post engagement using likes and comments.

The results show that Carousel posts received the highest average likes, while Reel posts received the highest average comments in this dataset. These findings demonstrate how different post types can generate different forms of audience engagement.

The analysis provides a basic foundation for understanding Instagram content performance and can be extended further using engagement-rate calculations and machine learning techniques.

8. Project Deliverable

GitHub Repository:

<https://github.com/dudebhuvanasai/Alfido-Tech-Internship>

Notebook:

Alfido_Instagram_Data_Analysis.ipynb